

```
Processando config 1, seed 3, teste: BlockAlphabit
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
```

```
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
```

```
Correction factor of the ChiSquare:
```

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 0.35
p-value of test : 0.84
```

```
-----
CPU time used : 00:00:00.01
```

```
Generator state:
```

```
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
```

```
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 3.93
p-value of test : 0.86

CPU time used : 00:00:00.02

Generator state:

33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 142.01
p-value of test : 0.19

CPU time used : 00:00:00.01

Generator state:

50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33290.08
p-value of test : 0.02

CPU time used : 00:00:00.02

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 8594.61
p-value of test : eps *****

CPU time used : 00:00:00.02

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305	
Chi-square statistic	:	3140.16	
p-value of test	:	eps	*****

CPU time used : 00:00:00.01

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-0.26	
p-value of test	:	0.60	

CPU time used : 00:00:00.01

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32	
ChiSquare statistic	:	880.94	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	31	
ChiSquare statistic	:	1871.62	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	450.29	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	59.95	
p-value of test	:	1.1e-4	*****

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	46.95	
p-value of test	:	3.7e-5	*****

CPU time used : 00:00:00.08

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	579.43	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	519.62	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	268.13	
p-value of test	:	1.8e-7	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	44.30	
p-value of test	:	0.89	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	20.06	
p-value of test	:	0.95	

CPU time used : 00:00:00.06

Generator state:
150994944 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.31
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):

(eps1 means a value < 1.0e-15):

	Test	p-value
5	HammingIndep, L = 16	eps
6	HammingIndep, L = 32	eps
8	RandomWalk1 H (L = 64)	eps
8	RandomWalk1 M (L = 64)	eps
8	RandomWalk1 J (L = 64)	eps
8	RandomWalk1 R (L = 64)	1.1e-4
8	RandomWalk1 C (L = 64)	3.7e-5
9	RandomWalk1 H (L = 320)	eps
9	RandomWalk1 M (L = 320)	eps
9	RandomWalk1 J (L = 320)	1.8e-7

All other tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 1.19
p-value of test                   : 0.55
```

```
-----
CPU time used                     : 00:00:00.01
```

Generator state:
167772160 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 13.65
p-value of test : 0.09

CPU time used : 00:00:00.01

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 154.74
p-value of test : 0.05

CPU time used : 00:00:00.01

Generator state:
201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32500.60
p-value of test : 0.85

CPU time used : 00:00:00.03

Generator state:
218103808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----
Number of degrees of freedom      : 177
Chi-square statistic              : 172.60
p-value of test                   : 0.58
-----
```

```
-----
CPU time used                     : 00:00:00.01
-----
```

Generator state:
234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

```
-----
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0
-----
```

Counters with expected numbers ≥ 10

```
-----
Number of degrees of freedom      : 305
Chi-square statistic              : 287.10
p-value of test                   : 0.76
-----
```

```
-----
CPU time used                     : 00:00:00.00
-----
```

Generator state:
251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 32
-----
```

```
-----
Normal statistic                  : -0.33
p-value of test                   : 0.63
-----
```

CPU time used : 00:00:00.01

Generator state:
268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.82
p-value of test	:	0.73

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	37.42
p-value of test	:	0.20

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	29.93
p-value of test	:	0.57

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	18.04
p-value of test	:	0.84

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	19.66
p-value of test	:	0.19

CPU time used : 00:00:00.07

Generator state:
285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	981.20	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	410.54	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	249.16	
p-value of test	:	8.1e-6	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	50.37	
p-value of test	:	0.72	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	24.88	
p-value of test	:	0.81	

CPU time used : 00:00:00.05

Generator state:
301989888 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.26
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	8.1e-6
-----	-----

All other tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 3.56
p-value of test : 0.17

CPU time used : 00:00:00.02

Generator state:
318767104 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 2.52
p-value of test : 0.96

CPU time used : 00:00:00.01

Generator state:
335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216

Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 133.43
p-value of test : 0.35

CPU time used : 00:00:00.01

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32610.25
p-value of test : 0.73

CPU time used : 00:00:00.02

Generator state:
369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 167.84
p-value of test : 0.68

CPU time used : 00:00:00.02

Generator state:

385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 6214.23
p-value of test : eps *****

CPU time used : 00:00:00.00

Generator state:

402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.17
p-value of test : 0.43

CPU time used : 00:00:00.01

Generator state:

419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 2752.93
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 788.29
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 197.63
p-value of test : eps *****

Test on the values of the Statistic R

Number of degrees of freedom : 25
ChiSquare statistic : 31.29

p-value of test : 0.18

Test on the values of the Statistic C

Number of degrees of freedom : 15
ChiSquare statistic : 18.99
p-value of test : 0.21

CPU time used : 00:00:00.06

Generator state:
436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom : 62
ChiSquare statistic : 997.09
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 62
ChiSquare statistic : 421.12
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 160
ChiSquare statistic : 334.19
p-value of test : 2.3e-14 *****

Test on the values of the Statistic R

Number of degrees of freedom : 57
ChiSquare statistic : 61.22
p-value of test : 0.33

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 39.86
p-value of test : 0.16

CPU time used : 00:00:00.06

Generator state:
452984576 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.27
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
6 HammingIndep, L = 32	eps
8 RandomWalk1 H (L = 64)	eps
8 RandomWalk1 M (L = 64)	eps
8 RandomWalk1 J (L = 64)	eps
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	2.3e-14

All other tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 2
Value of the statistic       : 0.25
p-value of test              : 0.88
```

CPU time used : 00:00:00.01

Generator state:

469761792 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic       : 18.14
p-value of test              : 0.02
```

CPU time used : 00:00:00.00

Generator state:

486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 161.75
p-value of test : 0.02

CPU time used : 00:00:00.02

Generator state:

503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32723.49
p-value of test : 0.57

CPU time used : 00:00:00.01

Generator state:
520093440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 12566.17
p-value of test : eps *****

CPU time used : 00:00:00.01

Generator state:
536870656 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 5462.28
p-value of test : eps *****

CPU time used : 00:00:00.00

Generator state:
553647872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.22
p-value of test : 0.59

CPU time used : 00:00:00.01

Generator state:
570425088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 2899.45
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 1424.40
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	331.81	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	24.80	
p-value of test	:	0.47	

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	20.13	
p-value of test	:	0.17	

CPU time used : 00:00:00.06

Generator state:
587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	1588.63	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	624.42	
p-value of test	:	eps	*****

```
-----
Test on the values of the Statistic J

Number of degrees of freedom      : 160
ChiSquare statistic               : 310.17
p-value of test                   : 1.1e-11      *****
```

```
-----
Test on the values of the Statistic R

Number of degrees of freedom      : 57
ChiSquare statistic               : 80.67
p-value of test                   : 0.02
```

```
-----
Test on the values of the Statistic C

Number of degrees of freedom      : 32
ChiSquare statistic               : 55.48
p-value of test                   : 6.2e-3
```

```
-----
CPU time used                     : 00:00:00.06
```

```
Generator state:
603979264 bits have been read.
```

===== Summary results of Alphabit =====

```
Version:          TestU01 1.2.3
Generator:        ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:   16777216
Number of statistics: 17
Total CPU time:   00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):
```

Test	p-value
5 HammingIndep, L = 16	eps
6 HammingIndep, L = 32	eps
8 RandomWalk1 H (L = 64)	eps
8 RandomWalk1 M (L = 64)	eps
8 RandomWalk1 J (L = 64)	eps
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	1.1e-11

```
-----
All other tests were passed
```

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
```

XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	5.87
p-value of test	:	0.05

CPU time used : 00:00:00.00

Generator state:

620756480 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 13.94
p-value of test : 0.08

CPU time used : 00:00:00.01

Generator state:
637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 149.17
p-value of test : 0.10

CPU time used : 00:00:00.01

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32765.66
p-value of test : 0.50

CPU time used : 00:00:00.02

Generator state:

671088128 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 169.52
p-value of test : 0.64

CPU time used : 00:00:00.01

Generator state:

687865344 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 370.69
p-value of test : 5.9e-3

CPU time used : 00:00:00.00

Generator state:

704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.75
p-value of test : 0.23

CPU time used : 00:00:00.01

Generator state:

721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	43.70
p-value of test	:	0.08

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	28.85
p-value of test	:	0.58

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	24.10
p-value of test	:	0.84

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	25.29
p-value of test	:	0.45

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	8.87
p-value of test	:	0.88

CPU time used : 00:00:00.06

Generator state:
738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

```

-----
Test on the values of the Statistic H

Number of degrees of freedom      :    62
ChiSquare statistic               : 1815.46
p-value of test                   :    eps      *****

```

```

-----
Test on the values of the Statistic M

Number of degrees of freedom      :    62
ChiSquare statistic               :  991.73
p-value of test                   :    eps      *****

```

```

-----
Test on the values of the Statistic J

Number of degrees of freedom      :   160
ChiSquare statistic               :  266.54
p-value of test                   :  2.5e-7      *****

```

```

-----
Test on the values of the Statistic R

Number of degrees of freedom      :    57
ChiSquare statistic               :   69.03
p-value of test                   :    0.13

```

```

-----
Test on the values of the Statistic C

Number of degrees of freedom      :    32
ChiSquare statistic               :   47.13
p-value of test                   :    0.04

```

```

-----
CPU time used                     : 00:00:00.06

```

```

Generator state:
754973952 bits have been read.

```

===== Summary results of Alphabit =====

```

Version:          TestU01 1.2.3
Generator:        ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:   16777216
Number of statistics: 17
Total CPU time:   00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

```

```

Test                                     p-value
-----

```

```
9 RandomWalk1 H (L = 320)      eps
9 RandomWalk1 M (L = 320)      eps
9 RandomWalk1 J (L = 320)      2.5e-7
```

All other tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen:  r = 0,    s = 32,    w = 32
```

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

```
Number of degrees of freedom      : 2
Value of the statistic             : 0.35
p-value of test                   : 0.84
```

CPU time used : 00:00:00.00

Generator state:

771751168 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen:  r = 0,    s = 32,    w = 32
```

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,

Sparse = FALSE

Number of bits = n = 16777216

Number of cells = 2^L = 16

Expected number per cell = 1048576

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 4.70
p-value of test : 0.79

CPU time used : 00:00:00.01

Generator state:

788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216

Number of cells = 2^L = 256

Expected number per cell = 65536

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 135.61
p-value of test : 0.31

CPU time used : 00:00:00.00

Generator state:

805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32629.82
p-value of test : 0.70

CPU time used : 00:00:00.02

Generator state:
822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 162.55
p-value of test : 0.77

CPU time used : 00:00:00.01

Generator state:
838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	281.71
p-value of test	:	0.83

CPU time used : 00:00:00.01

Generator state:
855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-0.14
p-value of test	:	0.55

CPU time used : 00:00:00.01

Generator state:
872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	25.78
p-value of test	:	0.77

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	20.49
p-value of test	:	0.92

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	25.42
p-value of test	:	0.79

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	28.75
p-value of test	:	0.27

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	13.92
p-value of test	:	0.53

CPU time used : 00:00:00.06

Generator state:
889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32
```

swalk_RandomWalk1 test:

```
-----
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320
```

Test on the values of the Statistic H

```
Number of degrees of freedom      : 62
ChiSquare statistic                : 1710.40
p-value of test                   : eps      *****
```

Test on the values of the Statistic M

```
Number of degrees of freedom      : 62
ChiSquare statistic                : 823.04
p-value of test                   : eps      *****
```

Test on the values of the Statistic J

```
Number of degrees of freedom      : 160
ChiSquare statistic                : 290.10
p-value of test                   : 1.4e-9    *****
```

Test on the values of the Statistic R

```
Number of degrees of freedom      : 57
ChiSquare statistic                : 57.53
p-value of test                   : 0.46
```

Test on the values of the Statistic C

```
Number of degrees of freedom      : 32
ChiSquare statistic                : 34.90
p-value of test                   : 0.33
```

CPU time used : 00:00:00.06

Generator state:
905968640 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	1.4e-9
-----	-----

All other tests were passed